USN | | | | | | | | | | [22IOTDE11]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

DATA WAREHOUSING AND DATA MINING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	What are the functions performed by OLAP?	L1	CO1	**02**
	b.	What do you understand by fact table?	L2	CO2	**02**
	c.	Define decision tree.	L2	CO3	**02**
	d.	Discuss about prediction.	L2	CO4	**02**
	e.	Define Clustering in Data Mining.	L2	CO4	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain in detail about ROLAP Server Architecture.	L2	**CO1**	**05**
		OR			
	b.	Compare OLTP and OLAP.	L2	**CO1**	**05**
3.	**a.**	Explain in detail about Data mining functionalities.	L2	**CO2**	**05**
		OR			
	b.	List out OLAP operations in multidimensional data model with diagrams.	L1	**CO2**	**05**
4.	**a.**	Write in brief about Data cleaning.	L2	**CO3**	**05**
		OR			
	b.	What is Data reduction? Discuss in detail.	L2	**CO3**	**05**
5.	**a.**	Illustrate Bayesian belief network in detail.	L3	**CO4**	**05**
		OR			
	b.	Describe Bayes theorem with example.	L2	**CO4**	**05**
6.	**a.**	What are the basic approaches for generating an agglomerative hierarchical clustering?	L2	**CO5**	**05**
		OR			
	b.	Explain k-Means and k-Medoids partitioning methods in detail.	L2	**CO5**	**05**
7.	**a.**	Discuss about world wide web (WWW) and how to mint the WWW.	L3	**CO6**	**05**
		OR			
	b.	Discuss about Multimedia data mining.	L2	**CO6**	**05**
		PART - C			**6x10=60**
8.	**a.**	Draw and explain the three tier data warehouse architecture.	L2	**CO1**	**10**
		OR			
	b.	Discuss about the Star schema, Snow Flake schema, and Fact constellation schema.	L3	**CO1**	**10**

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9.	**a.**	Explain the architecture of data mining.	L2	**CO2**	**10**
		OR			
	b.	Explain different data pre- processing techniques.	L2	**CO2**	**10**
10.	**a.**	Explain about the Mining Multilevel Association rules with example.	L2	**CO3**	**10**
		OR			
	b.	Describe the steps involved in improving the efficiency of the Apriori algorithm.	L2	**CO3**	**10**
11.	**a.**	How does the Naïve Bayesian classification works? Explain in detail.	L2	**CO4**	**10**
		OR			
	b.	Explain decision tree induction algorithm for classifying data tuples and with suitable example.	L2	**CO4**	**10**
12.	**a.**	Discuss in detail about Partitioning methods in clustering with examples.	L2	**CO5**	**10**
		OR			
	b.	What is clustering analysis? Explain different types of data in clustering with an example.	L2	**CO5**	**10**
13.	**a.**	What is Web Mining? Different between Web Content Mining, Web Structure Mining and Usage Mining.	L2	**CO6**	**10**
		OR			
	b.	What are the social impacts of Data Mining?	L2	**CO6**	**10**